

Private Key Compromise & Multi-Chain Fund Flow Investigation

Project Overview

This case study examines a private key compromise incident involving unauthorized asset transfers across multiple blockchain environments, including BNB Chain, TRON, and Bitcoin network infrastructure.

The investigation reconstructed transaction continuity from the initial unauthorized wallet access through intermediary wallet movement, cross-chain conversion behaviour, stablecoin routing, aggregation activity, bridge interaction, and downstream Bitcoin peeling-chain movement.

The case demonstrates practical blockchain investigation methodology involving compromise-response tracing, cross-chain analysis, behavioural transaction assessment, and Bitcoin UTXO reconstruction.

Wallet addresses, transaction hashes, screenshots, bridge identifiers, and visual references have been partially masked in this public portfolio version for professional presentation and responsible disclosure considerations.

INVESTIGATION SUMMARY

Category		Details
Investigation Type		Private Key Compromise & Multi-Chain Fund Flow Investigation
Networks Reviewed		BNB Chain → TRON → Bitcoin
Assets Reviewed		BNB, USDT, BTC
Investigation Focus		Unauthorized Fund Movement & Cross-Chain Laundering Analysis
Observed Behaviour		Layering, bridge routing, aggregation, Bitcoin peeling-chain movement
Final Status		Multiple downstream BTC holding wallets identified
Key Observations		
- Rapid wallet-draining activity identified - Newly created intermediary wallets observed - Cross-chain bridge interaction detected - Stablecoin conversion and aggregation behaviour identified - Bitcoin peeling-chain structure reconstructed		

INVESTIGATION OBJECTIVES & DASHBOARD

Investigation Objectives

- Analyse unauthorized transaction movement following wallet compromise
- Review intermediary wallet behaviour
- Assess cross-chain bridge interaction activity
- Examine stablecoin conversion behaviour
- Analyse aggregation and consolidation patterns
- Conduct Bitcoin peeling-chain analysis
- Identify downstream holding-wallet activity
- Demonstrate structured blockchain investigation methodology

Investigation Dashboard

Initial Incident

Private key compromise suspected

Unauthorized Activity

Immediate outbound wallet draining identified

Layering Behaviour

Intermediary forwarding wallets observed

Cross-Chain Activity

BNB → TRON bridge routing identified

Bitcoin Activity

Structured BTC peeling-chain behaviour identified

Final Destination

Multiple downstream BTC holding wallets observed

Methodology:

The investigation combined blockchain transaction tracing, bridge analysis, wallet interaction review, stablecoin flow assessment, and Bitcoin UTXO analysis to reconstruct the movement of compromised funds across multiple blockchain environments.

METHODOLOGY

Methodology Applied

- Unauthorized transaction tracing
- Wallet interaction analysis
- Cross-chain bridge review
- Stablecoin routing analysis
- Fund consolidation assessment
- Timestamp correlation
- Behavioural transaction analysis
- Bitcoin UTXO tracing
- Peeling-chain analysis
- Aggregation behaviour review
- Manual transaction flow reconstruction

Tools & Platforms Referenced

Tools referenced in this investigation varied depending on the blockchain network and analytical requirements of the case.

Platforms Used

BscScan

TronScan

Blockchain.com Explorer

Mempool.space

Bridge transaction explorers

Arkham Intelligence

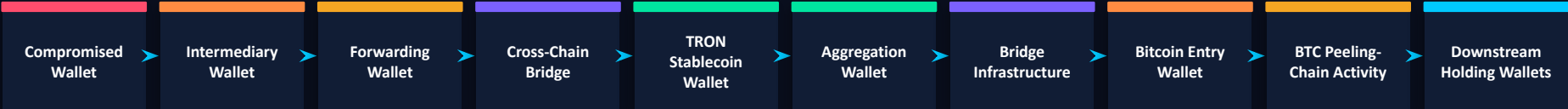
Breadcrumbs

OKLink

Bitcoin transaction graph visualisation tools

MULTI-CHAIN TRANSACTION RECONSTRUCTION

The reconstructed transaction sequence begins from the compromised wallet before progressing through intermediary wallets, bridge infrastructure, TRON-based stablecoin routing, aggregation behaviour, and downstream Bitcoin peeling-chain movement.



Masked Transaction References

Type	Masked TXID	Status
Initial Unauthorized Transfer	0x91ab3d...f72c81	Confirmed
Bridge Transaction	0x4ec7a1...91dd52	Confirmed
Stablecoin Routing	7f3ad8...d119e0	Confirmed
BTC Transfer	a91d72...4e88bc	Confirmed

Masked Wallet References

Wallet Type	Masked Address
Compromised Wallet	0x8d20Eb...E0D479
Intermediary Wallet	0x41fd82...9f7A12
TRON Wallet	TNac9nV...HemJ
BTC Wallet	bc1qxy2k...f4k9

TRANSACTION FLOW ANALYSIS — STAGES 1–4

Stage 1 — Initial Unauthorized Wallet Activity

The investigation identified unauthorized outbound transfers originating from the compromised wallet shortly after the suspected private key exposure event.

Key Findings

- Immediate wallet-draining behaviour observed
- Newly created recipient wallets identified
- No prior meaningful wallet history detected

Assessment: The rapid outbound transaction behaviour demonstrated characteristics commonly associated with automated compromise exploitation and operational laundering preparation.

Stage 2 — Intermediary Forwarding Activity

The compromised funds were subsequently forwarded through additional intermediary wallets before cross-chain routing activity.

Key Findings

- Sequential forwarding behaviour observed
- Minimal holding duration identified
- Layered transaction routing detected

Assessment: The reviewed forwarding sequence demonstrated structured intermediary separation designed to reduce direct trace continuity from the compromised source wallet.

Stage 3 — Cross-Chain Bridge Interaction

The investigation identified bridge activity converting BNB-chain assets into TRON-based stablecoin infrastructure.

Key Findings

- BNB routed into bridge infrastructure
- Stablecoin conversion identified
- TRON destination wallet activity observed

Assessment: The bridge interaction represented a laundering transition point enabling movement between blockchain ecosystems and increasing tracing complexity.

Stage 4 — TRON Stablecoin Routing

Following bridge conversion, the funds were routed through TRON-based stablecoin wallets before further downstream movement.

Key Findings

- Stablecoin receipt activity identified
- Structured routing behaviour observed
- Additional intermediary movement detected

Assessment: The use of TRON stablecoin routing supported rapid low-cost movement and subsequent aggregation behaviour.

TRANSACTION FLOW ANALYSIS — STAGES 5–8

Stage 5 — Aggregation & Consolidation Behaviour

The investigation identified consolidation behaviour involving aggregation of multiple inflows into centralized wallet infrastructure.

Key Findings

- Multi-inflow aggregation identified
- Consolidated balances observed
- Structured routing sequence detected

Assessment: The reviewed aggregation behaviour demonstrated operational consolidation patterns commonly associated with laundering workflows involving staged routing and pooled liquidity handling.

Stage 6 — Bitcoin Entry Routing

Following stablecoin routing activity, the funds were bridged into Bitcoin network infrastructure.

Key Findings

- Bitcoin entry wallet identified
- Stablecoin-to-BTC conversion behaviour observed
- Downstream UTXO routing initiated

Assessment: The Bitcoin transition introduced a UTXO-based tracing environment involving peeling-chain reconstruction and downstream transaction analysis.

Stage 7 — Bitcoin Peeling-Chain Analysis

The investigation identified structured Bitcoin peeling-chain behaviour involving sequential partial transfers across multiple wallets.

Key Findings

- Repeated output splitting observed
- Sequential downstream transfers identified
- Residual balance retention behaviour observed
- Multiple intermediary UTXO transitions detected

Assessment: The observed Bitcoin transaction structure demonstrated characteristics commonly associated with peeling-chain laundering methodology designed to incrementally distribute and obscure transaction flow continuity.

Stage 8 — Final Holding Wallets

The final observed routing stage involved multiple downstream Bitcoin holding wallets retaining unspent balances.

Key Findings

- Multiple downstream holding wallets identified
- Residual BTC balances remained unspent
- No confirmed centralized exchange interaction observed during reviewed flow

Assessment: The retained balances suggest temporary storage or delayed cash-out behaviour pending future movement.

EVIDENCE ASSESSMENT — CORE EVIDENCE PILLARS

The analytical conclusions are supported through multiple independent indicators identified across all reviewed blockchain environments.

Continuous Multi-Chain Transaction Flow

Transaction continuity was preserved across: BNB Chain, bridge infrastructure, TRON stablecoin routing, and Bitcoin network movement.

Layered Routing Behaviour

Multiple intermediary wallets introduced operational separation between: compromised wallet activity, bridge routing, aggregation behaviour, and Bitcoin movement.

Stablecoin Aggregation Activity

TRON-based stablecoin routing demonstrated structured consolidation and pooled transaction behaviour prior to Bitcoin conversion.

Bitcoin Peeling-Chain Behaviour

The Bitcoin transaction structure demonstrated sequential UTXO peeling patterns involving: repeated transfers, partial residual retention, and downstream wallet propagation.

SCREENING & EXPOSURE REVIEW

Screening Observations

- Cross-chain bridge interaction identified
- Stablecoin aggregation behaviour observed
- Multi-stage layering activity detected
- Bitcoin peeling-chain methodology identified
- Multiple downstream holding-wallet relationships observed

Behavioural Risk Indicators

Risk Indicator	Assessment
Private Key Compromise Behaviour	High
Intermediary Wallet Layering	High
Cross-Chain Laundering Activity	High
Bridge Infrastructure Usage	High
Bitcoin Peeling-Chain Behaviour	High
Residual Holding Wallet Structure	Medium

Screenshots & Visual Evidence — Included Evidence Types

Blockchain explorer screenshots · Cross-chain bridge confirmations · Wallet interaction diagrams · Transaction timeline visuals · Bitcoin UTXO flow diagrams · Peeling-chain mapping visuals · Aggregation flow structures

Portfolio Safety Controls: masked wallet addresses · masked TXIDs · masked screenshots where appropriate · shortened visual identifiers · reduced exposure of sensitive identifiers

INVESTIGATION SCOPE, LIMITATIONS & ANALYTICAL ASSESSMENT

Investigation Scope & Limitations

This investigation was conducted using publicly accessible blockchain data, bridge transaction analysis, wallet interaction review, and open-source analytical techniques.

Wallet ownership, beneficial ownership, and downstream identity attribution could not be independently verified using public blockchain data alone.

No privileged exchange records, KYC-linked account information, or law-enforcement investigative access were available during the review.

Analytical findings are therefore based on:

- Observable transaction behaviour
- Transaction continuity
- Cross-chain routing analysis
- Bridge interaction review
- Stablecoin aggregation behaviour
- Bitcoin UTXO analysis
- Peeling-chain assessment

Analytical Assessment

The reviewed activity demonstrated a structured multi-chain laundering sequence following a suspected private key compromise involving:

- Intermediary wallet layering
- Bridge infrastructure usage
- Stablecoin routing
- Aggregation behaviour
- Bitcoin peeling-chain movement
- Downstream holding-wallet retention

The cross-chain transitions significantly increased tracing complexity while preserving sufficient transaction continuity for behavioural reconstruction across all reviewed blockchain environments.

The Bitcoin peeling-chain behaviour represented one of the strongest laundering indicators identified during the investigation.

CONCLUSION

Conclusion

The investigation successfully reconstructed transaction continuity across multiple blockchain networks and identified behavioural indicators associated with layered laundering activity and structured fund movement following a suspected private key compromise.

The case demonstrates how compromised digital assets can be routed through multiple blockchain ecosystems and structured transaction methodologies to increase tracing complexity while preserving operational liquidity movement.

The reviewed activity demonstrated:

Rapid Wallet-Draining Behaviour

Intermediary Wallet Separation

Bridge-Enabled Cross-Chain Routing

Stablecoin Aggregation Behaviour

Bitcoin Peeling-Chain Methodology

Downstream Holding-Wallet Retention

Recommended Next Steps

Based on the reconstructed transaction flow and identified downstream wallet relationships, continued monitoring of the associated wallets and intermediary infrastructure is recommended.

- Ongoing Bitcoin wallet monitoring
- Expansion of peeling-chain analysis
- Monitoring for centralized exchange interaction
- Review of future bridge-routing behaviour
- Preservation of transaction evidence and timeline records
- Continued sanctions and exposure screening

Final Project Summary

This project demonstrates practical blockchain investigation methodology involving:

**Compromise-Response
Tracing**

Multi-Chain Analysis

**Bridge Transaction
Review**

**Stablecoin Routing
Assessment**

**Wallet Interaction
Analysis**

Bitcoin UTXO Tracing

**Peeling-Chain
Reconstruction**

**Behavioural Transaction
Analysis**

**Structured AML-Focused
Reporting**

The investigation combined blockchain analysis, cross-chain tracing, behavioural assessment, and Bitcoin transaction reconstruction techniques to map the movement of compromised funds across multiple blockchain ecosystems.